ENGINEERING & ENERGY HACKATHON 2026

Challenge problem statements & data sources.

Three tracks, mapped to the sponsors' strategic objectives, each underpinned by real, downloadable open data and a clear policy hook.

Start, grow and thrive in Energy & Sustainability.

SPONSORED BY

UKERC UK Energy Research Centre · SUPERGEN ENERGY NETWORKS · SuperAIRE · EDRC Energy Demand Research Centre

Event: 11 September 2026 · Edgbaston Conference Centre, University of Birmingham.

Draft v5 · incorporates the partner-academic review of v3 and the SuperAIRE review of Track 2 · 25 August 2026

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01 · OVERVIEW

What this is

Draft problem statements for the three confirmed tracks (A-10). Each is written to the partner steer: underpinned by a real, named, open dataset; a clear DESNZ policy hook; grounded in real demographic or energy-access data where appropriate; and scoped to a single working day. Draft v4 incorporates the partner-academic review of v3 (comments received 12 and 13 August); the accompanying change summary maps every comment to its change.

How these map to the sponsors' strategy

The hackathon is sponsored by UKERC, Supergen Energy Networks, SuperAIRE and EDRC. The strategic objectives the tracks serve are shared across that sponsor group; the wording is quoted from the UKERC 2024-2029 strategy. The three tracks address all four challenges - Affordability & Justice (T1), Flexibility and Delivery (T2), Delivery plus Affordability & Justice (T3), and the Geopolitical Challenge through a security-of-supply and resilience lens carried in T2. We reference DESNZ policy as the real-world hook; we do not claim a formal sponsor-DESNZ partnership.

DESIGN CONSTRAINTS

- One in-person day; the team working session is roughly 4 to 5 hours. Each challenge is tractable in that window with data handed to teams.
- Mixed discipline and tooling (engineering, social science, policy, data). Every task is attemptable in spreadsheet or Power BI as well as Python.
- EDI by design: plain-language briefs, clear outputs, roles for non-coding disciplines on every track.
- Capacity building is a scored outcome (see judging and the data-and-reproducibility standard).
- Teams may bring published literature and their own domain evidence; the staged data is the floor, not the ceiling.

CHALLENGE TRACK

02 Energy Poverty & Equity

STRATEGIC OBJECTIVE · AFFORDABILITY & JUSTICE

"understanding and mitigating any negative distributional impacts of the energy transition on households and wider society."

POLICY HOOK

The Warm Homes Plan. LILEE and the Warm Home Discount are England's measurement and support instruments; the devolved nations define and measure fuel poverty differently.

> In which small areas (LSOAs) does fuel-poverty risk concentrate most once you account for both household vulnerability and poor housing efficiency, and where would a fixed programme budget, a fund the size of a typical support scheme, cut that risk most?

THE TASK · HALF-DAY

1. Join sub-regional fuel-poverty statistics to a deprivation layer at LSOA / LA level.
2. Build a simple, explainable, weighted index ranking small areas by combined need.
3. Pick one intervention from the provided menu (one fabric measure, one bill-support measure; indicative cost and effect parameters, including the solid-wall caution, in the pre-event pack), show which areas to fund first and why.
4. Show the distributional consequence: who is helped, who is missed, what the index is blind to.
5. Would the people targeted take this up? Name the acceptance and delivery barriers (trust, stigma, landlord consent) your allocation ignores.

DATA · OPEN, DIRECT DOWNLOAD, OGL

- DESNZ sub-regional fuel poverty 2026 (2024 data, LILEE, LSOA) - gov.uk/government/statistics/sub-regional-fuel-poverty-data-2026-2024-data
- English Indices of Deprivation 2019 (IMD, File 7, join on LSOA) - gov.uk/government/statistics/english-indices-of-deprivation-2019
- NEED 2024 (property-level, 50k sample) - gov.uk/government/statistics/national-energy-efficiency-data-framework-need-anonymised-data-2024

Scope the anchor files (LILEE fuel poverty, English IMD) cover England, and NEED covers England and Wales, so the on-the-day analysis is England-only. A team extending to the devolved nations starts from the devolved sources listed on the sources page rather than hunting for data. Interventions come from the provided menu with provided parameters.

Roles data (join + index) · social science (weighting + what it hides) · policy (delivery mechanism) · engineering (intervention assumptions). **Output** 5-min pitch + notebook/spreadsheet + honest limitations. **Stretch** a second weighting scenario to show targeting sensitivity.

CHALLENGE TRACK

03 Future Electricity Systems

STRATEGIC OBJECTIVE · FLEXIBILITY, DELIVERY + GEOPOLITICAL

"understanding the role of new sources of flexibility necessary for the decarbonisation of heat, power and transport." Also carries the Geopolitical Challenge (geopolitics, security and resilience) through a security-of-supply lens: flexibility and renewables reduce exposure to imported gas and interconnector dependence at peak.

POLICY HOOK

Clean Power 2030 Action Plan and the NESO clean-power pathway. Matches SuperAIRE and Supergen Energy Networks.

> Using real GB electricity data, how much carbon (and cost) could a defined flexible load save by shifting when it runs, and what would it take to unlock that flexibility?

THE TASK · HALF-DAY

1. Take a recent period of carbon-intensity and demand data. All three system sources are half-hourly (settlement-period) resolution; analyse at least one full recent week, staged in the pre-event pack so every team starts from the same window. A winter-versus-summer week contrast is a good stretch.
2. Model one flexible load: an electricity use whose timing can move within a stated window without losing the service it provides (EV charging that must finish by morning, heat-pump pre-heating, or a batch process). State its size (kWh per run) and its shiftable window as explicit assumptions.
3. Compute the carbon difference between worst-time and best-time running, minimising carbon intensity (cost as a stretch).
4. Ground it in one named local authority using the sub-national consumption data; explore which types of signal, tariff or incentive could enable the shift, and the opportunities and challenges of each. Any incentive recommendation is an illustrative policy proposal under stated assumptions, not a prediction of actual consumer behaviour.
5. Who cannot flex and why (tenure, occupancy, control over appliances), and what makes a tariff acceptable rather than merely rational? Name the equity risk of rewarding those who can flex.

Tractability the shift model is a sort-and-sum over the intensity series - spreadsheet-tractable; a worked starter is in the pre-event pack.

> **Security-of-supply lens · Geopolitical Challenge**
>
> The generation mix already shows gas (CCGT) and interconnector imports. Import reliance here means the share of GB generation met at a given time by interconnector imports plus gas-fired generation, the two sources most exposed to imported fuel and cross-border markets, both visible in the mix data. Show how the flexible load's shift reduces that reliance at peak - the resilience dimension of a highly renewable system. No extra data needed.

DATA · OPEN, NO API KEY

- Carbon Intensity API (national + regional, no key) - api.carbonintensity.org.uk/intensity
- NESO Historic Generation Mix + Historic Demand Data - neso.energy/data-portal
- Elexon Insights (BMRS), generation by fuel + demand - bmrs.elexon.co.uk
- DESNZ sub-national electricity consumption (place / equity) - gov.uk/government/collections/sub-national-electricity-consumption-data

Staged data the staged datasets ship with a one-page data dictionary (fields, units, resolution, period, source, licence) and a baseline-assumptions sheet (default load size, shiftable window, worked local-authority example).

Scope the system analysis is GB-wide: the Carbon Intensity, NESO and Elexon datasets cover the GB grid (Northern Ireland operates within the all-island Single Electricity Market). The place lens is one named GB local authority from the sub-national consumption data, with the Carbon Intensity API's regional endpoint available for a regional cut. The policy context is UK-wide.

Roles engineering/data (model + optimisation) · data/AI (forecast the window - SuperAIRE angle) · policy (incentive/tariff) · social science (who can and cannot flex). **Output** 5-min pitch + model. **Note** wholesale price is only partially open, so cost stays a stretch.

04 CHALLENGE TRACK
Heat, Buildings & Decarbonisation

STRATEGIC OBJECTIVE · DELIVERY + AFFORDABILITY & JUSTICE

"managing the massive scale of new physical assets and capital needed to decarbonise the energy system", with the justice lens on cold, damp homes.

POLICY HOOK

Awaab's Law (Phase 1 for social landlords in England from 27 October 2025) - a proactive duty to investigate and act on damp and mould within statutory timeframes; plus the Warm Homes Plan and clean-heat rollout. The EoH demonstration was itself a DESNZ programme, delivered by Energy Systems Catapult.

Framing · the right ruler, and evidence-led retrofit

A home can pass on one method and fail on another. If the assessment is short in the unsafe direction, a landlord can log a compliant survey and still miss the hazard - directly relevant to Awaab's Law. Recommend evidence-led sequencing: measure, control, prove it held, then size fabric to measured need - not model-first retrofit.

> Across real heat-pump homes, which are most at risk of condensation and mould on cold surfaces - and does the answer change when you assess with monthly averages (as the condensation standard's Glaser method does) versus measured overnight lows?

THE TASK · HALF-DAY

1. For each property, compute a surface-condensation margin on the coldest day (formula and worked method in the pre-event pack).
2. Compare two views of the same homes: monthly-average (Glaser) vs measured overnight-minimum. Count how many flip from "fine" to "at risk".
3. Name the assumptions (indoor humidity is not measured; external-climate basis matters) and show which moves the result most.
4. Turn it into a decarbonisation-and-equity point: who gets wrongly reassured, and what should an assessor or retrofit scheme do differently?
5. What does an at-risk verdict mean for the resident and for the landlord's duty, and what would make acting on it feasible?

What is and is not claimed

Claimed: on this cohort, temporal resolution can change the verdict, under-reporting overnight risk - a method result that reproduces on public EoH data. **Not claimed:** UK-wide prevalence or any causal health claim (the cohort is small, self-selected, no measured humidity). Pair any equity conclusion with representative data; work at cohort level and never turn residents into the data product.

WAYS TO EXTEND

The two-view comparison is the required spine; beyond it, teams may take the data further: which property types flip; sensitivity to the external-climate basis; propose your own risk metric; check humidity against the IDEAL dataset. Teams whose extensions show promise are invited to continue the analysis in follow-on research after the event - noting honestly that the underlying data is England-only at the moment.

DATA

- EoH Performance Summary (ESC/DESNZ, account-free; per-property coldest-day temps) - opennetzero.org › electrification-of-heat interim performance data summary

- MCS Data Dashboard (installs by area) - datadashboard.mcscertified.com · DESNZ sub-national gas (heating-demand proxy) · NEED 2024 (fabric/EPC band).
- Pre-download (login): EPC bulk register (GOV.UK One Login) - get-energy-performance-data.communities.gov.uk
- Stretch only: IDEAL Household Energy Dataset (measured room temperature and humidity, 255 UK homes; open) - datashare.ed.ac.uk/handle/10283/3647

Output 5-min pitch + the analysis: at-risk counts under each view, the assumption that moves the result most, and the assessment recommendation. Worked-example method is in the pre-event pack; reproduction repo not yet public.

05 · SHARED

Judging, standard & traceability

JUDGING CRITERIA · DRAFT, FOR A-13

- Insight and relevance to the challenge and its policy hook -25%.
- Correct, defensible use of the real data, assumptions stated -25%.
- Reproducibility and transparency: re-runnable, no hardcoded headline, honest limitations -20%.
- Interdisciplinary collaboration across engineering, data, social science, policy -15%.
- Communication: a plain-language pitch a non-specialist policymaker could act on -15%.

DATA-AND-REPRODUCIBILITY STANDARD · CAPACITY BUILDING

- Record where each dataset came from and when; verify any provided file against its published hash.
- Never hardcode a headline number; compute it from the data in front of the judges.
- State every assumption as an assumption; show which one the result is most sensitive to.
- Prefer an explainable method a policymaker can trust over an opaque one.

THE ARC · STRATEGIC OBJECTIVE TO DELIVERABLE TO OUTCOME

TRACK	STRATEGIC OBJECTIVE	PROBLEM	TEAM DELIVERABLE	OUTCOME FOR SPONSORS
1	Affordability & Justice	Fuel-poverty targeting index + allocation	Pitch + index + funding recommendation	An illustrative, explainable approach to equity targeting - a method demonstration, not a deployable targeting tool - plus capacity in fuel-poverty data
2	Flexibility, Delivery + Geopolitical	Carbon/cost of shifting a load; import reliance	Pitch + flexibility model + incentive	Quantified flexibility value and security-of-supply insight
3	Delivery + Aff. & Justice	Does the method change the damp verdict	Pitch + two-view at-risk analysis	Assessment-method finding relevant to Awaab's Law and retrofit sizing

Anchor data and adequacy are set per track (see each track and Data readiness). The arc closes beyond the day: capacity built, collaborations formed, and policy-relevant outputs are captured in the post-event impact report (A-20) and the six-month follow-up (A-22), feeding the sponsors' shared mission to inform government and build interdisciplinary capability.

06 · DATA READINESS

Everything staged on the Drive

Actions A-2 / A-11. Links checked live on 16 July 2026. Nothing is left to download on the day: every dataset is staged to the shared Google Drive ahead of the event, so teams open a folder, not a browser. Only genuinely large datasets are staged as a sample or recent window, and live APIs are staged as a snapshot with the live endpoint kept for anyone wanting fresh data.

DATASET	SIZE	STAGED ON THE DRIVE AS
DESNZ fuel poverty; IMD 2019; NEED 2024 sample; DESNZ sub-national gas/electricity	Small - moderate	Full files (MSOA where the LSOA file is large)
NESO generation mix & demand; Elexon BMRS	Large / live	A recent snapshot (e.g. last 12 months); live endpoint noted
Carbon Intensity API	Live	A snapshot for the challenge period; live endpoint noted
EoH Performance Summary	Small	Verified copy + SHA256
EPC register	Massive (~30M certs)	A per-local-authority CSV subset, not the bulk
IDEAL Household Energy Dataset (stretch)	Large, open (Edinburgh DataShare)	Relevant files staged (subset if needed); accept the licence

Open items before this is final

Co-refine each track with a partner academic (A-10) and confirm exact dataset extracts (A-11). Confirm: the EoH Performance Summary licence; current-year figures (fuel-poverty vintage, Clean Power 2030) at pack time. Publish the reproduction repo before circulating any repo link to teams.

07 · SOURCES

Sources & verification

Self-verification

Every source below carries a specific link, checked live on 16 July 2026. Figures are attributed to named public sources; where a widely-cited figure comes from a secondary analysis it is flagged at source. Claims are bounded - see Track 3, "what is and is not claimed". Working draft for partner refinement, not a formal audit.

STRATEGY & POLICY

- UKERC 2024-2029 strategy - ukerc.ac.uk/about/strategy
- Clean Power 2030 Action Plan - gov.uk/government/publications/clean-power-2030-action-plan
- NESO clean-power pathway - neso.energy/publications/clean-power-2030
- Awaab's Law: guidance for social landlords (in force 27 Oct 2025) - gov.uk/government/publications/awaabs-law-guidance-for-social-landlords

DATASETS

- DESNZ sub-regional fuel poverty 2026 - gov.uk/government/statistics/sub-regional-fuel-poverty-data-2026-2024-data
- DESNZ sub-national electricity consumption - gov.uk/government/collections/sub-national-electricity-consumption-data
- DESNZ sub-national gas consumption - gov.uk/government/collections/sub-national-gas-consumption-data
- DESNZ sub-national total final energy consumption - gov.uk/government/collections/total-final-energy-consumption-at-sub-national-level
- English Indices of Deprivation 2019 - gov.uk/government/statistics/english-indices-of-deprivation-2019
- NEED anonymised data 2024 - gov.uk/government/statistics/national-energy-efficiency-data-framework-need-anonymised-data-2024
- Carbon Intensity API - api.carbonintensity.org.uk · NESO data portal - neso.energy/data-portal · Elexon Insights (BMRS) - bmrs.elexon.co.uk
- EoH Performance Summary (ESC / DESNZ) - opennetzero.org › electrification-of-heat interim performance data summary
- MCS Data Dashboard - datadashboard.mcscertified.com · EPC bulk register (GOV.UK One Login) - get-energy-performance-data.communities.gov.uk
- IDEAL Household Energy Dataset (measured room temperature and humidity, 255 homes; open, licence acceptance) - datashare.ed.ac.uk/handle/10283/3647 (DOI 10.7488/ds/2836)
- EoH temporal-averaging reproduction (RGBY.ai) - not currently public; the method is in the pre-event pack

DEVOLVED-NATIONS STARTING POINTS (TRACK 1 EXTENSION; ADDED IN V4, LIVE-CHECK AT PACK TIME)

- Scotland: Scottish House Condition Survey (fuel-poverty statistics) - gov.scot/collections/scottish-house-condition-survey · SIMD 2020 - gov.scot/collections/scottish-index-of-multiple-deprivation-2020
- Wales: fuel-poverty estimates - gov.wales/fuel-poverty-estimates-wales · WIMD 2019 - gov.wales/welsh-index-multiple-deprivation
- Northern Ireland: NI House Condition Survey - nihe.gov.uk › house-condition-survey · NI Multiple Deprivation Measure - nisra.gov.uk/statistics/deprivation

SPONSORS

- UKERC - ukerc.ac.uk · Supergen Energy Networks - supergenen.org · SuperAIRE - superaire.org · EDRC - edrc.ac.uk

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